AI & GENERATIVE AI TRACK / PROJECT 04 OF 05

LearnMate

PHASE 01 OF 08 | BRAINSTORMING & IDEATION

Project overview

LearnMate helps students work with scattered course materials through a REST API for uploads and AI-generated learning resources. The brief describes secure accounts, MongoDB storage, Gemini-powered content generation, and a dashboard-facing backend for saved study resources.

Project description

A study-support API for uploading learning materials and generating summaries, flashcards, quizzes, and personalized study plans.

Project status

Not stated in the source brief. Update this document when the team records an agreed status.

Project capabilities

- Study material uploads
- AI summaries and flashcards
- Practice quizzes
- Personalized study plans

Phase description

Define the user problem, explore ideas, and agree which opportunity is worth solving.

Project-specific phase focus

Map students' study-material and exam-preparation challenges into a clear learning need.

Phase status and progress

Status: Not stated in the source material.
Progress: Not recorded in the source material.

Tasks and requirements

[ ] List and group solution ideas
[ ] Prioritize ideas by feasibility and importance
[ ] Write the customer problem statement
[ ] Map what the user says, thinks, does, and feels

Relevant documents and data

Project brief: LearnMate - AI assistant for smarter learning and study support..docx

Phase templates:
- Brainstorming & Idea Prioritization (1. Brainstorming & Ideation/Brainstorming & Idea Prioritization.pdf)
- Define Problem Statements (1. Brainstorming & Ideation/Define Problem Statements .pdf)

- Empathy Map (1. Brainstorming & Ideation/Empathy Map.pdf)

Project technology stack

- Node.js
- Express.js
- MongoDB
- Mongoose
- Gemini AI

Phase notes and decisions